



CHEM Project

**Advanced Decision Support System for
Chemical/Petrochemical Manufacturing Processes**

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TB 3.5 User Manual

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1 MITforRD

1.1 Introduction

This part of the document describes the off-line part of Toolbox 3.5 software “Process modelling using Fuzzy Logic and Neural Networks for Process Assessment” intended to models building. In the rest of the manual this part of TB 3.5 will be called MITforRD.

The goal of this Toolbox is to generate residuals for fault detection and isolation purpose (utilized by TB5.5 or other TB), using Partial Parametric Models (PPM, see glossary). Residuals are generated through simple comparison of modelled and measured process variables. The basic entities that toolbox works with are PPMs. TB 3.5 also contains procedures for PPMs identification.

The main function of MITforRD is to obtain PPMs of process variables. The PPM is identified using given raw data (historical process data). In MITforRD software this task is divided to following

3 stages which correspond to software modules:

- Data Preprocessing – this module is used to prepare identification data from raw txt files. The following functionality is implemented in this module:
 - data importing from any txt files,
 - data saving and loading in internal MITforRD format,
 - data analysing: autocorrelation and power spectrum,
 - data preprocessing: filtering and trends removing.
- Models – provide all routines needed to model identification, also implement MIMO – MISO framework to models encapsulation. The following functionality is implemented:
 - configuration and identification of following PPMs models:
 - Classic Linear,
 - Polynomial,
 - Multi–Layer Preceptron,
 - Fuzzy Neural Network,
 - TSK Fuzzy Model,
 - Mendel–Wang Fuzzy Model.
 - MIMO – MISO models management.
- Verification – this module is responsible for model verification and validation based on provided data in MITforRD internal format (raw data prepared by Data Preprocessing module).

The architecture of MITforRD is based on plug-ins system. Functionality of each module can be extended or restrict by adding or removing plug-ins which are independent components of MITforRD. All functionality, with exception of simple GUI, is provided by plug-ins system e.g. all models configuration, identification routines and GUI are implemented via separated plug-ins. plug-ins are specially prepared dll libraries according to programming interface characteristic for each module: Data Preprocessing, Model Building and Verification. Also third party plug-ins can be added to MITforRD.

1.2 Common functionality – main window

1.2.1 Starting the Program

Starting of MITforRD depend of localization of whole TB3.5 software. In default case, after installing TB3.5 software, MITforRD can be launched by choosing shortcut from menu **Start | Programs | DSDiag | MITforRD**.

Other way to launch MITforRD is starting MITforRD.exe from working directory e.g. C:\Program Files\DSDiag\MITforRD.exe.

After luncthing the program the main window should opens.



Figure 1: MITforRD main window

Menu Bar – allow access to all function of the program by clicking corresponding menu item.

Toolbar – fast access to main functions of MITforRD via buttons.

Menu Bar

File Menu

- Plugin manager – invokes Plugin manager window; this window allow view information about each plug-in and Enables/Disables it,
- Show config – NO AVAILABLE,
- Exit – closes the program,

Modules Menu

- Data preprocessing – invokes Data preprocessing window; the raw data importing and preprocessing,
- Models – invokes Models window; models configuration and identification,
- Verification – invokes Verification window; models verification,
- Calculation manager – invokes Calculation manager window.

View Menu

- Messages – invokes Messages window; MITforRD event (errors, warnings info) viewer,

Help Menu (UNDER CONSTRUCTION)

Toolbar



– invokes Data preprocessing window,



– invokes Models window,



– invokes Verification window,



– invokes Messages window,



– invokes Plugin manager window,



– closes the program,

1.2.2 Plugin Manager window

The Plugin Manager window display information about each plugin.

Plugins are binary files with extension *.mbplug.

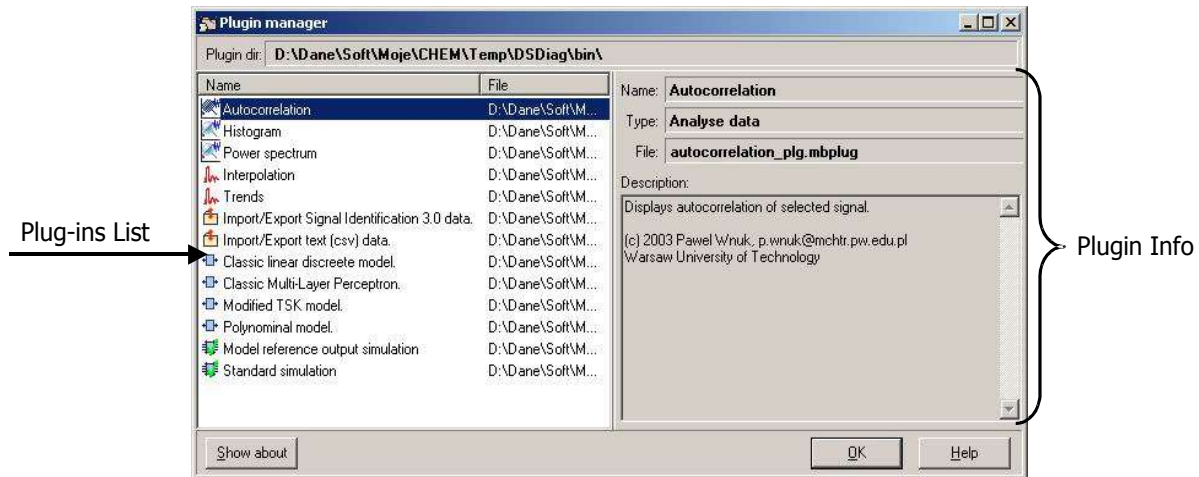


Figure 2: Plugin Manager window

Plugin dir text box show current directory from plug-ins were loaded.

Plug-ins List

Plug-ins list show in the form of the list all currently loaded plugins to MITforRD. The list is divided into two columns, **Name** column show name of plug-in and icon indicating type of plug-in. The **File** column display plugin's file name with path.

Plug-ins' type icons:



– analyse plugin,



– model plugin,



– preprocessing plugin,



– model verification plugin,



–import data plugin,

Plug-ins Info

Plug-ins info panel show information about currently selected plugin on plugin list. **Name** text box show the name of plugin, the **Type** is numerical ID of plugin's type and **File** show name of plugin's file name. Description text box show description for plugin e.g. authors, copyrights.

Show about button display About window of selected plugin.

Ok button closes the window.

1.2.3 Messages window – MITforRD events viewer

Messages window show MITforRD events, errors warnings and information. It's useful tool to trace how MITforRD work and detect any abnormal software behaviour.

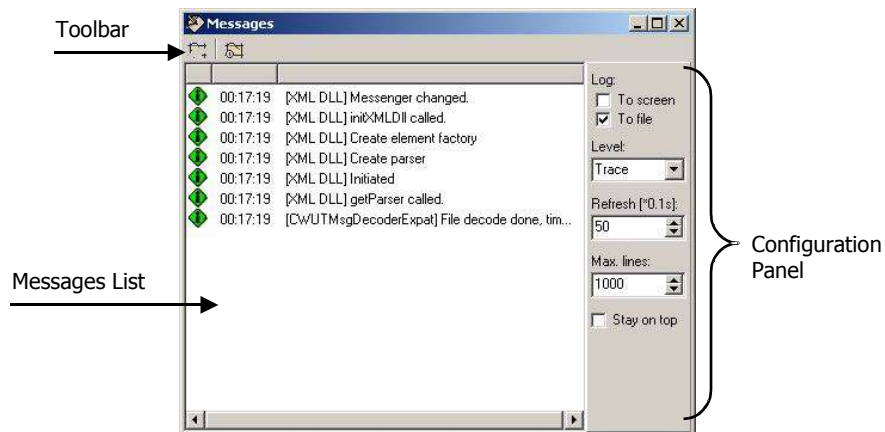


Figure 3: Messages window

Toolbar



– clear up messages list,



– closes the Messages window,

Messages List

Message list contain messages notified by MITforRD during it running. The list consist of three columns. First indicates using icon type of message, second one, notify time. Third column show source of message (in square brackets) with description.

Configuration Panel

Configuration panel allow adjust functionality of Messages window.

Log To screen check box enable/disable logging notified messages to messages list.

Log To file check box allow enable/disable logging notified messages to log file MITforRD.log (placed in working directory).

Level combo box allow filtering incoming messages by its types. The following types of messages are available: No Log, Errors, Warnings, Info, Debug and Trace (in order of message priority).

Refresh edit box allow set refresh time (in 100ms) of Messages window.

Max lines edit box is used to set capacity (in lines) of messages list.

Stay on top check box enable/disable stay on top property of Messages window.

1.2.4 Calculation manager window

Calculation manager window show current state and configuration of MITforRD Distributed Calculation System - DCS. This window useful only for model plugins that using DCS.

State and configuration of MITforRD DCS is provided by different views of Calculation manager window, Threads view, Local system load view and Thread times view. To change view click on view's corresponding button on View Selector.

Threads view

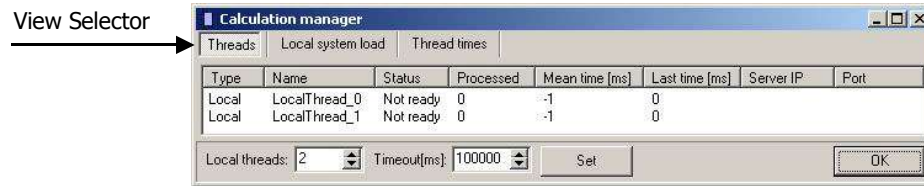


Figure 4: Calculation manager window – Threads view

Threads view show current state of all threads (local and remote) which carry out distributed calculation.

Local system load view



Figure 5: Calculation manager window – Local system load view

The bars on Local system load view mean:

- green bar – every processor load,
- yellow bar – total system load

Thread times view

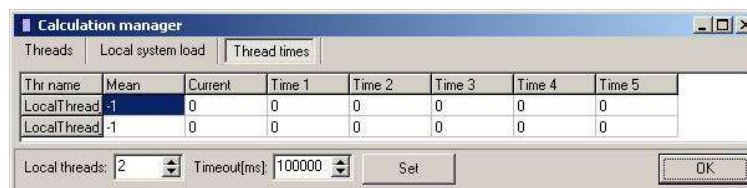


Figure 6: Calculation manager window – Thread times

This view is completed by each model plugins which utilize MITforRD DCS.

1.3 Data preprocessing

1.3.1 Invoking Data preprocessing window

To invoke Data preprocessing window, choose Modules | Data preprocessing from Main Window menu bar, or click Data preprocessing icon on Main Window toolbar.

1.3.2 Data preprocessing window

Data Preprocessing window is main window of Data Preprocessing Module. It have capability to import data from any txt file and save it to internal MITforRD data format (it depend on currently loaded data import plugins). Also Preprocessing window allow to visualize data in the form of plot or table and analysing it using analysing plug-ins. Data can be modified by hand using table or using specialized preprocessing plug-ins (e.g. trends removing).

The data processed by Data Preprocessing Module are time series of real process data, e.g. temperature, tank level, pressure. Values of process variables in different time period (time series) will be called signals, set of signals will be called data or data set.

Data Preprocessing window allow display data and its properties in several views – Data editor view, Plots view, Details view and Signals details view.

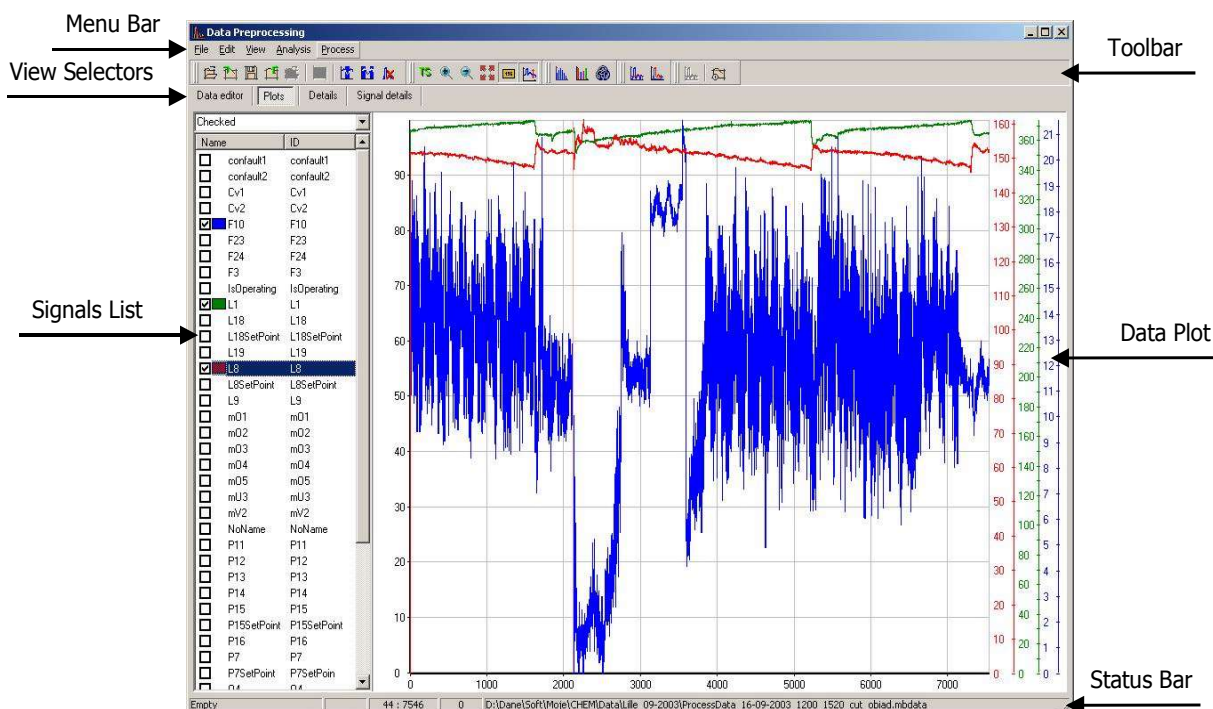


Figure 7: Data Processing window – Plots view

Menu Bar

File Menu

- Load – load data from file in MITforRD data format,
- Save – save data to file in MITforRD data format; user can choose between two formats, binary which is much faster to load (*.mbbin) and text format (*.mbdata) which is human readable,
- Close – close data set,
- Import data – invokes Select import filter plugin window (see 1.3.2.5 paragraph),
- Export data – invokes Select import filter plugin window (see 1.3.2.5 paragraph),

Exit – closes the window,

Edit Menu

Cut – invokes Cut part of data window; cut a part of data set; user should provide First sample and Last sample of cut part of data (see figure 8); all samples before First sample are deleted as well as any samples after Last sample,

Delete – invokes Delete part of data window; delete a part of data set; user should provide First sample and Last sample of deleted part of data (see figure 8),

Delete signal – delete current signal from data set,

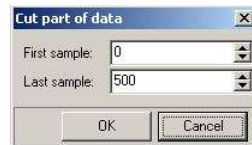


Figure 8: Cut data window

View Menu

Show time stamps – enable/disable showing time stamps on Plots view,

Zoom in – zoom in time axis of Plots view,

Zoom out – zoom out time axis of Plots view,

Undo zoom – undo zoom and full plot of Plots view,

Normalize – enable/disable plots normalize on Plots view,

Analysis Menu

This menu depend on set of installed plugins (see Analysis plugins chapter).

Process Menu

This menu depend on set of installed plugins (see Preprocessing plugins chapter).

Toolbar

-  – load data from file in MITforRD data format
-  – import data from txt files or other file formats,
-  – save data to file in MITforRD data format
-  – export data to txt files or other file formats,
-  – invokes Cut part of data window
-  – invokes Delete part of data window,
-  – delete current signal from data set,
-  – enable/disable showing time stamps on Plots view,
-  – zoom in time axis of Plots view,
-  – zoom out time axis of Plots view,
-  – undo zoom and full plot of Plots view,

-  – enable/disable showing of cursor position on Plots view,
-  – enable/disable plots normalize on Plots view,
-  – close the window

Toolbox appearance depend on which plugins are loaded to MITforRD.

View Selector

View selector buttons are used to switch between views. To change view click on view's corresponding button.

The Data editor view allow display data in tab form. This view can be used to edit data.

The Plots view is used to visualize data in plot form. One signal or multiple signals can be displayed simultaneously.

The Details view show basic information about data set such us name, sampling time and description.

The Signals details view is used to configure properties of signals.

Status Bar

Status bar show basic information about current data set. Each fields of status bar are described on Figure 9.

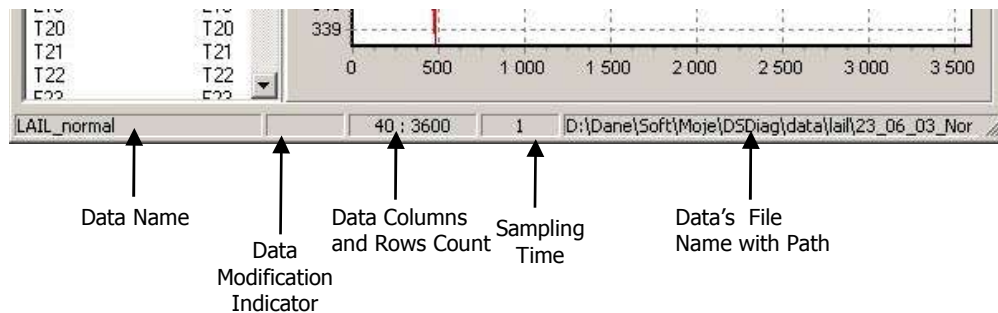
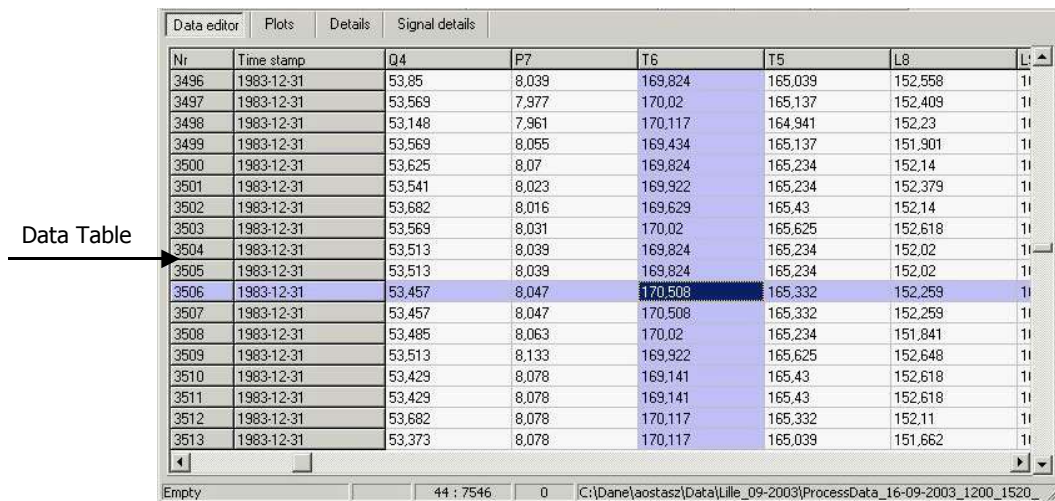


Figure 9: Status bar of Data Processing window

1.3.2.1 Data editor view

To enable data editor view press Data editor button on View Selector.

The data editor view is used to display and edit data in table form. Data can be edited and saved to file, it is useful to remove e.g. outliers.



Nr	Time stamp	Q4	P7	T6	T5	L8	
3496	1983-12-31	53,85	8,039	169,824	165,039	152,558	11
3497	1983-12-31	53,569	7,977	170,02	165,137	152,409	11
3498	1983-12-31	53,148	7,961	170,117	164,941	152,23	11
3499	1983-12-31	53,569	8,055	169,434	165,137	151,901	11
3500	1983-12-31	53,625	8,07	169,824	165,234	152,14	11
3501	1983-12-31	53,541	8,023	169,922	165,234	152,379	11
3502	1983-12-31	53,682	8,016	169,629	165,43	152,14	11
3503	1983-12-31	53,569	8,031	170,02	165,625	152,618	11
3504	1983-12-31	53,513	8,039	169,824	165,234	152,02	11
3505	1983-12-31	53,513	8,039	169,824	165,234	152,02	11
3506	1983-12-31	53,457	8,047	170,508	165,332	152,259	11
3507	1983-12-31	53,457	8,047	170,508	165,332	152,259	11
3508	1983-12-31	53,485	8,063	170,02	165,234	151,841	11
3509	1983-12-31	53,513	8,133	169,922	165,625	152,648	11
3510	1983-12-31	53,429	8,078	169,141	165,43	152,618	11
3511	1983-12-31	53,429	8,078	169,141	165,43	152,618	11
3512	1983-12-31	53,682	8,078	170,117	165,332	152,11	11
3513	1983-12-31	53,373	8,078	170,117	165,039	151,662	11

Figure 10: Data Processing window –Data editor view

Data Table

Data table allow show data in numeric form. Columns are representation of signals, column caption consist name of signal. Rows represent values of signal in different time periods. Row caption consist number of sample and time stamp, if is available.

Current position in Data table is represented by blue highlighted row and column. Bad samples are marked by red color, rows with at least one wrong samples are pink. Good samples on table are white.

To enable table's cell editor just click twice on cell.

1.3.2.2 Plots view

To enable Plots view press **Plots** button on View Selector.

The Plots view is used to visualize data in plot form. On plot can be displayed one signals or multiple signals. Plot's behavior can be adjusted by appropriate **Toolbar** buttons, for details see **Toolbar** description.

Signals List

Signal List is used to choose which signals should be plot. In the top of Signal List is combo box used to control list behavior, see Figure 11.

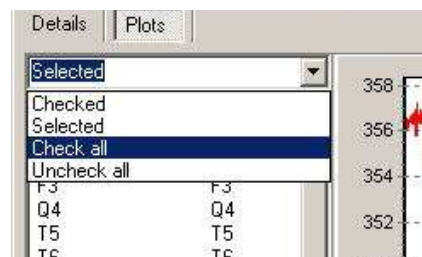


Figure 11: Signals List control combo box; Data Processing window - plot view

Changing behaviour of Signals List

- ◆ to change **Selected** list to **Checked** list (signals' list with check boxes), choose **Checked**, **Check all** or **Uncheck all** items of combo; this option allow to plot multiple signals on the same plot by marking signal on the list;
- ◆ to change **Checked** list to **Selected** list, choose **Selected** item of combo; this option allow to plot only one signals on plot by choosing it from the list;

- ◆ Check All item is used to mark all signals on list (combo will automatically change to Checked) – all signals are plotted;
- ◆ Uncheck All item allow to unmark all signals on list (combo will automatically change to Checked) – clears plot area;

1.3.2.3 Details view

To enable Details view press **Details** button on View Selector.

The Details view show basic information about data set such name, sampling time and description.

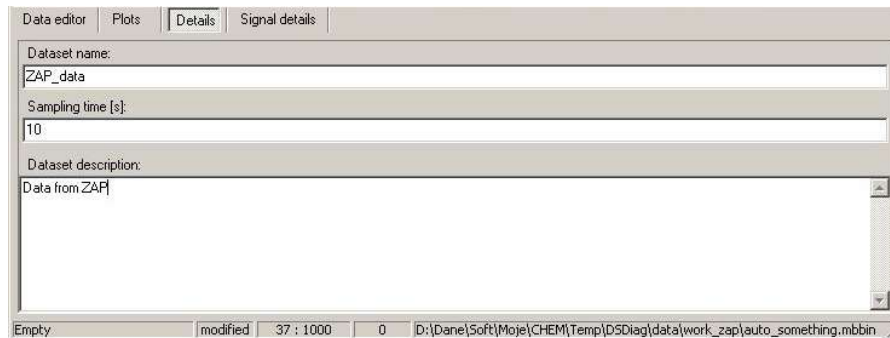


Figure 12: Data Processing window –Details view

1.3.2.4 Signal details view

To enable Signal details view press **Signal details** button on View Selector.

The Signals details view is used to configure properties of signals such as ID, Name, Range, Unit and Description.

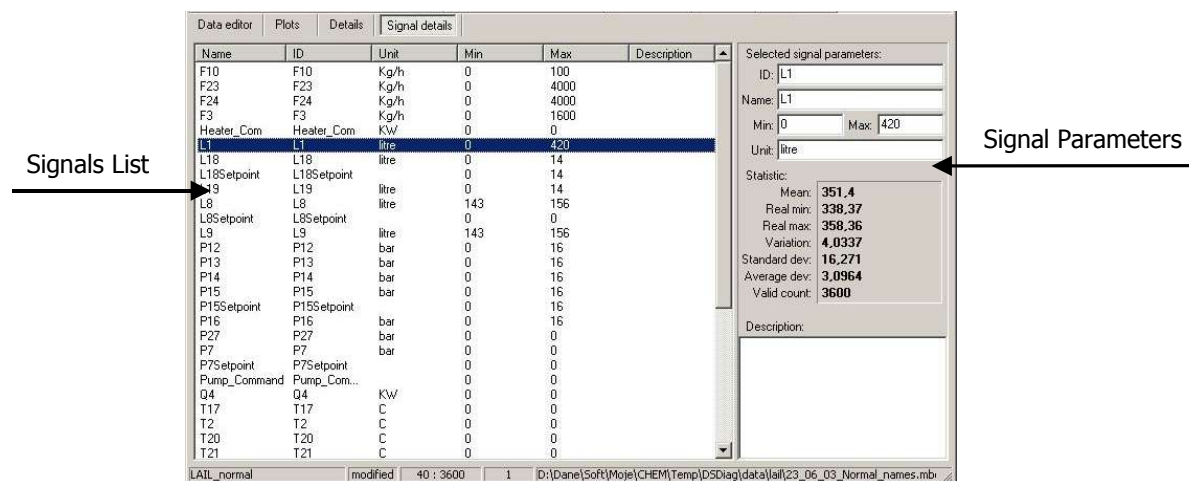


Figure 13: Data Processing window – Signal details view

Signals List

Signals List is used to choose signals, also it shows all properties of signals. To choose the signal click on corresponding row of the list. The signal's properties will appear on Signal Parameters panel.

Signal Parameters

Signal Parameters panel show parameters of single signal and can be used to edit it. The following properties can be edited:

ID – unique id of signal,

Name – name of signal

Min – min value of signal e.g. min of measure range,

Max – max value of signal e.g. max of measure range,

Unit – unit of signal e.g. m, m³/h,

Description – any description of signal.

The statistic part of Signal Parameters panel show basic statistical parameters of currently chosen signal.

1.3.2.5 Import data

To invoke Import Data option, choose File | Import data from menu bar of Data Preprocessing window menu bar, or click Import data icon on Data Preprocessing window toolbar.

Import data options of MITforRD is used to import data from different sources e.g. txt files. It's implemented as plugins.

After choosing Import Data option Select import filter window will appear (see Figure 14). Filters are plug-ins which are used to import data from different sources. Its availability depend on set of loaded Import plugins.

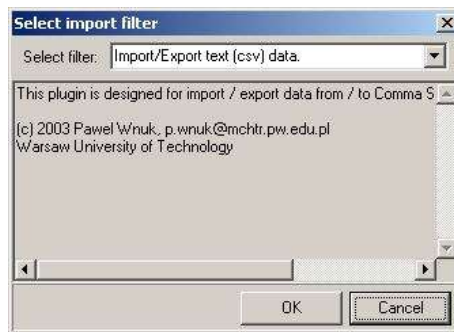


Figure 14: Import data – filter selection window

To choose filter, select it from **Select filter** combo box and click **Ok**.

For details on Import data plugins see Import data plugins chapter.

1.4 Models

1.4.1 MIMO – MISO framework description

1.4.2 Invoking Models window

To invoke Models window, choose Modules | Models from Main Window menu bar, or click Models icon on Main Window toolbar.

1.4.3 Models main window

Models window used as a main window of Models Module. It creates common environment to management of MIMO–MISO models. Also it provide all routines and GUI, via plug-in system, needed to PPMs configuration and identification.

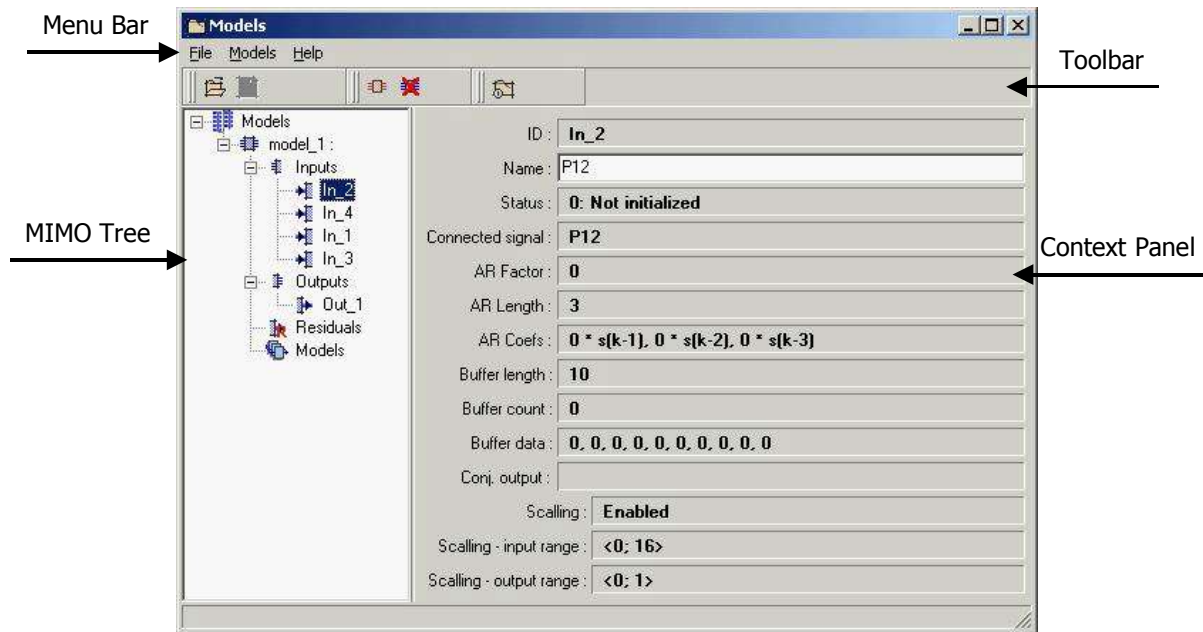


Figure 15: Models main window

The Models main window was designed with idea to keep all information about MIMO-MISO model in one place. Thus its behaviour depends on which option is chosen. Starting point for navigation via MIMO-MISO models structures is MIMO Tree. It is in the shape of a tree showing structures of all available MIMO and MISO models. By clicking on a tree item, the user can see its description on the Context Panel. The Context Panel always shows information of the currently chosen item of the MIMO Tree.

Menu Bar

File Menu

- Load – load MIMO model from file,
- Save – save MIMO model from file,
- Close – close the window,

Models Menu

- Create – invokes New MIMO Model window,

Help Menu (UNDER CONSTRUCTION)

Toolbar

-  – load MIMO model from file
-  – save MIMO model from file
-  – invokes **New MIMO Model** window
-  – delete selected MIMO models
-  – close the window

MIMO Tree

MIMO Tree is the starting point to navigation via MIMO-MISO models structures. It in shape of tree show structures of all available MIMO and MISO models.

To see description of each tree item, click on it. Description will be displayed on **Context Panel**.

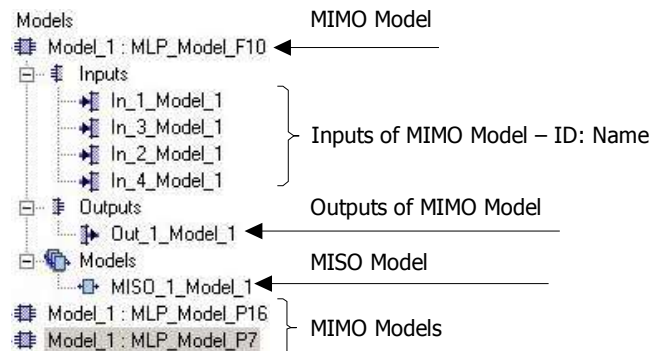


Figure 16: MIMO Tree of Models main window

MIMO Tree item's icons description:

-  – set of MIMO models,
-  – MIMO model,
-  – set of MISO models,
-  – MISO model,
-  – set of MIMO model inputs,
-  – input of MIMO model,
-  – set of MIMO model outputs,
-  – output of MIMO model.

Context Panel

Behaviour of Context Panel depend on currently chosen MIMO Tree item. The description of Context Panel will be presented for every it view.

MIMO Models view

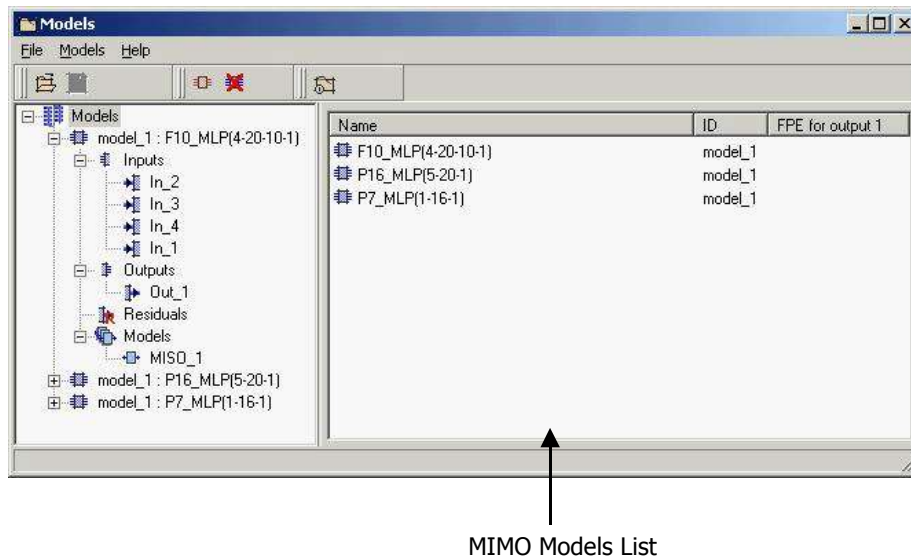


Figure 17: Models main window – MIMO models view

MIMO Model List show all currently loaded MIMO models. To select MIMO Model view, double click MIMO model item on the list.

MIMO Model view

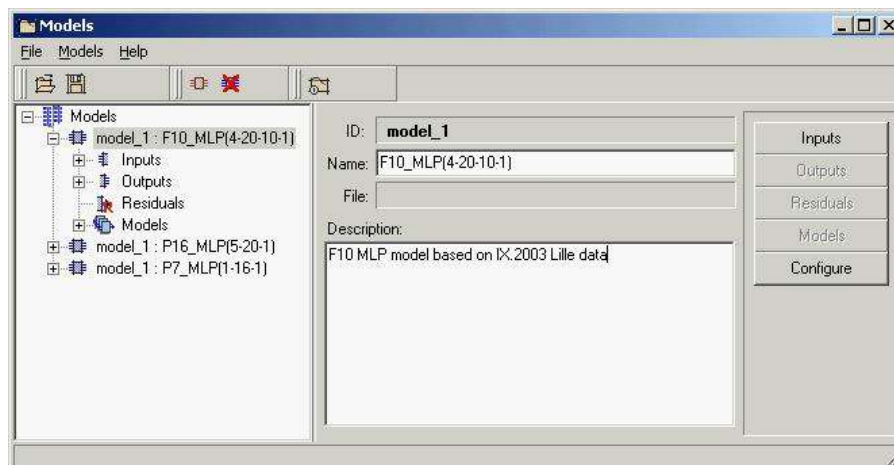


Figure 18: Models main window – MIMO model view

MIMO Model view display basic information about MIMO model.

To invoke MISO model configuration window click **Configure** button.

Inputs button switch Context Panel to MIMO Inputs List.

MIMO Inputs view

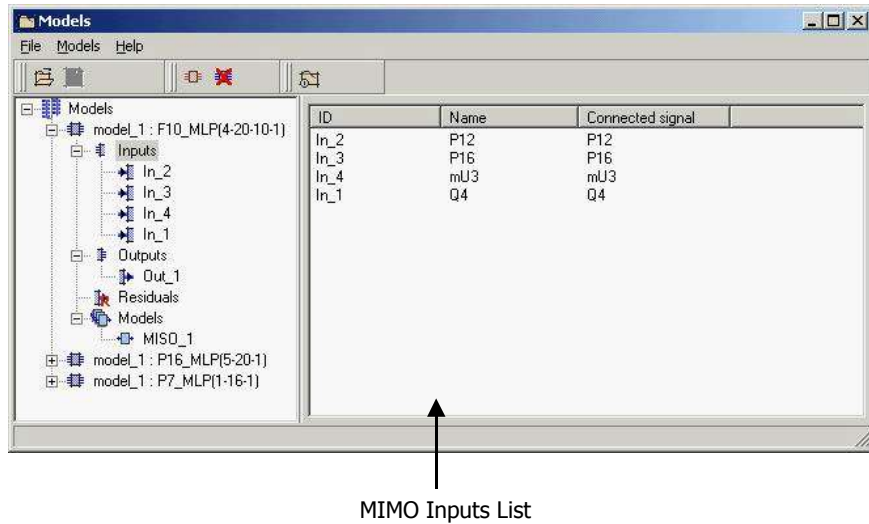


Figure 19: Models main window – MIMO inputs view

MIMO Inputs List show all inputs of MIMO model. To select MIMO input view, double click input item on the list (for column description see MIMO Input view paragraph below).

MIMO Input view

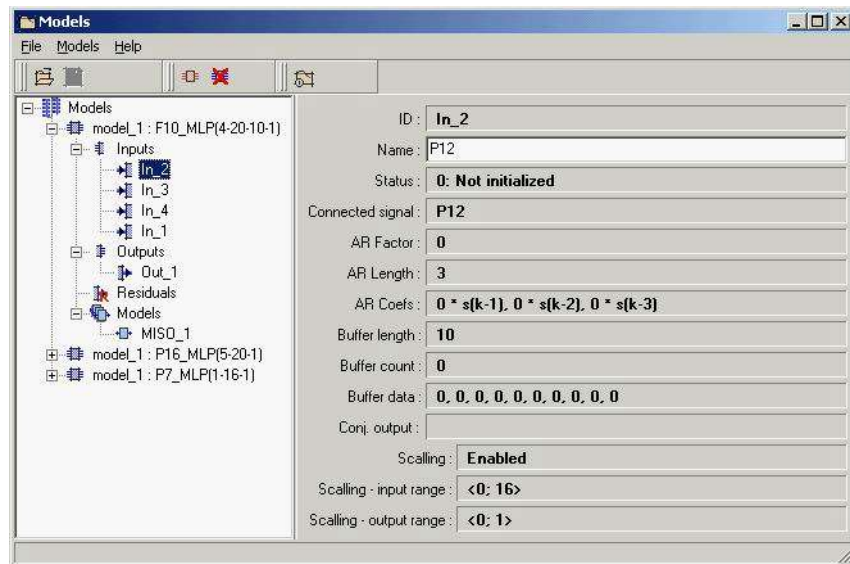


Figure 20: Models main window – MIMO input view

MIMO Input view show configuration of one input of MIMO model. This fields are provided for information purposes only, with exception of **Name** field which can be edit.

The MIMO Input are described by following parameters:

ID – unique input's ID,

Name – name of the input assigned by user,

Status – status of the input,

Connected signal – ID of the signal connected to the input (real time and historical values are taken from signal),

AR Factor, AR Length and AR coefs – parameters describes Auto Regressive model of the input (in fact it's AR model of signal),

Buffer length, Buffer count and Buffer data – parameters describes input's internal data buffer, length of the buffer, values count in buffer and its data,

Conj. output –

Scaling – indicate when scaling is Enabled/Disabled,

Scaling - input range, Scaling - output range – data provided by connected signal will be scaled from input range to output range.

MIMO Outputs view

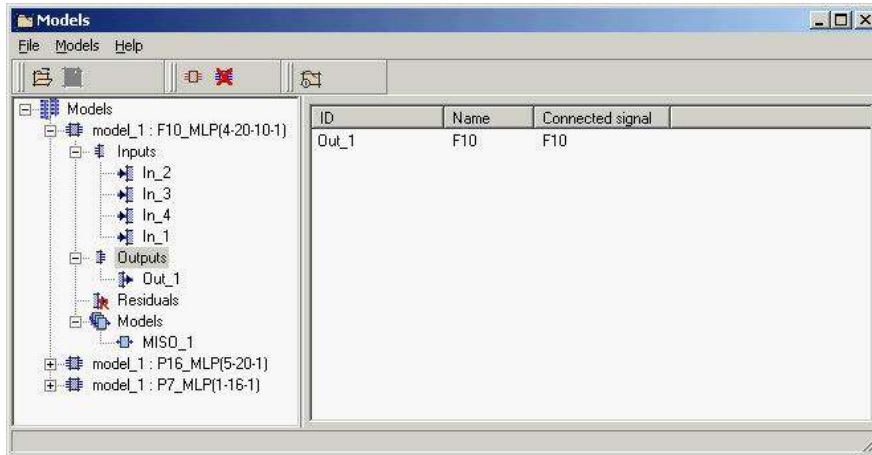


Figure 21: Models main window – MIMO outputs view

MIMO Outputs view show all outputs of MIMO model. To select MIMO output view, double click output item on the list (for column description see MIMO Output view paragraph below).

MIMO Output view

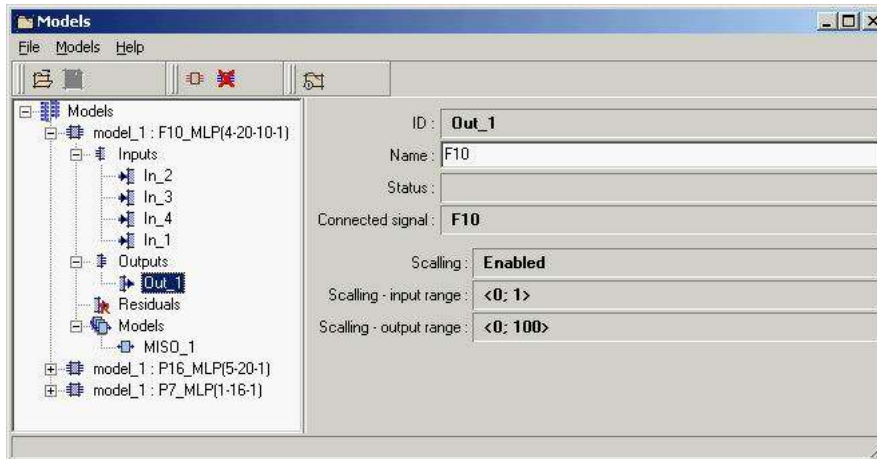


Figure 22: Models main window – MIMO output view

MIMO Output view show configuration of one output of MIMO model. This fields are provided for information purposes only, with exception of Name field which can be edit by the user.

The MIMO Output are described by following parameters:

ID – unique output's ID,

Name – name of the output assigned by user,

Status – status of the output,

Connected signal – ID of the signal connected to the output (calculated model's output will be assigned to this signal),

Scaling – indicate when scaling is Enabled/Disabled,

Scaling - input range, Scaling - output range – data put in the connected signal will be scaled from input range (model's output domain) to output range (connected signal domain).

MIMO Residuals view

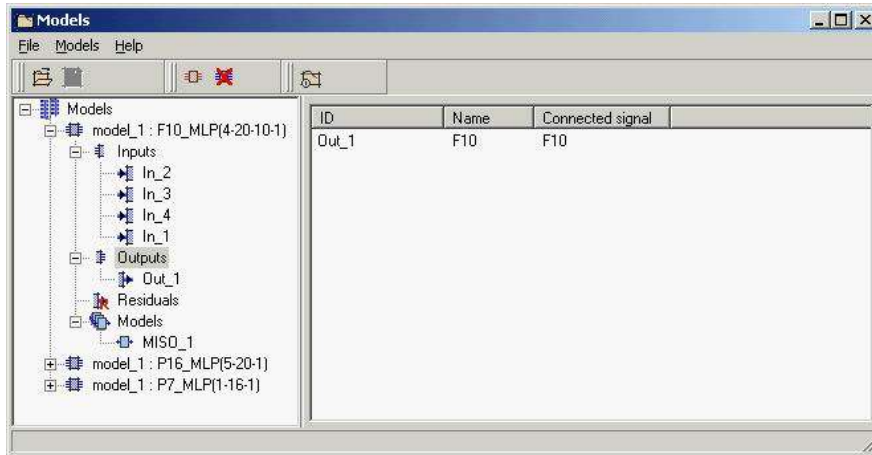


Figure 23: Models main window – MIMO residuals view

MIMO Residuals view show all residuals of MIMO model. To select MIMO Residual view, double click residual item on the list.

MIMO Residual view

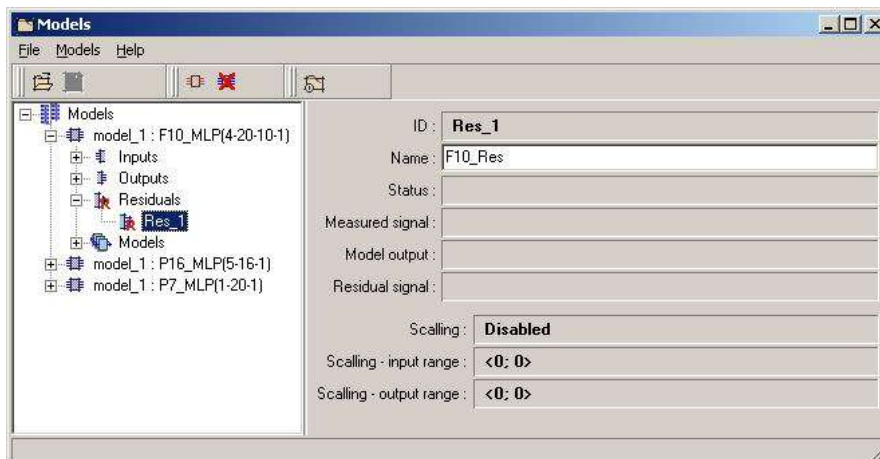


Figure 24: Models main window – MIMO output view

MIMO Residual view show configuration of one residual of MIMO model. This fields are provided for information purposes only, with exception of **Name** field which can be edit by the user.

The MIMO Residual are described by following parameters:

ID – unique residual's ID,

Name – name of the residual assigned by user,

Status – status of the residual,

Measured signal – ID of the signal for which residual is calculated,

Model signal – ID of the model's output used to calculate residual,

Residual signal – ID of the signal connected to the residual (calculated residual will be assigned to this signal),

Scaling – indicate when scaling is Enabled/Disabled,

Scaling - input range, Scaling - output range – data put in the connected Residual signal will be scaled from input range (residual's output domain) to output range (connected Residual signal domain).

MISO Models view

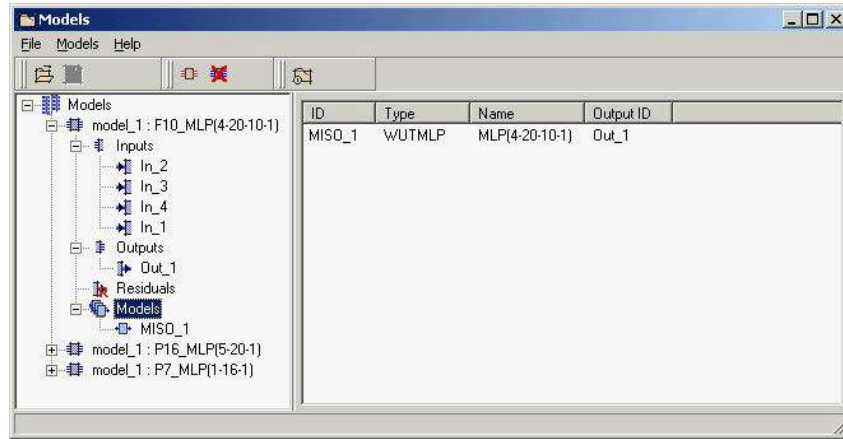


Figure 25: Models main window – MISO models view

MISO Models view show all MISO models of MIMO model. To select MISO model view, double click model item on the list.

MISO Model view

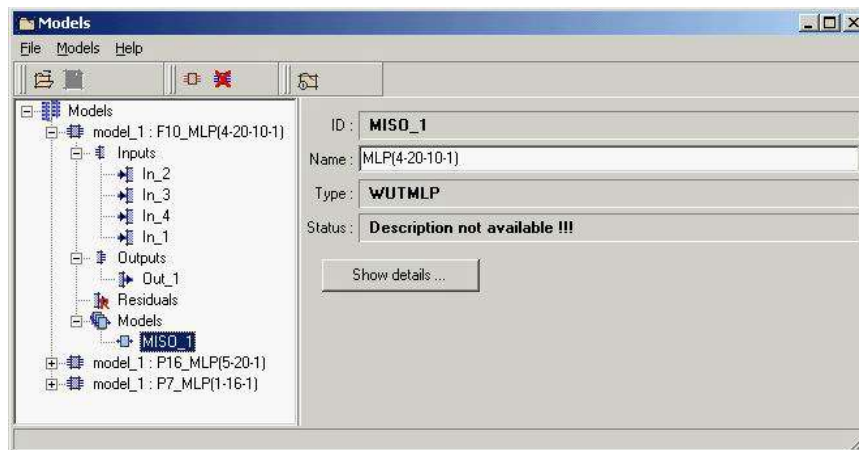


Figure 26: Models main window – MISO model view

MISO Model view show basic description of MISO model. This fields are provided for information purposes only, with exception of Name field which can be edit by the user.

The common parameters for all MISO Models are as follows:

ID – unique model's ID,

Name – name of the model assigned by user,

Type – model's type,

Status – status of the model,

More detailed description, model plugin depended, is available by Show details ... button.

1.4.4 MIMO configuration window

To invoke MIMO configuration window click **Configure** button from MIMO model view of **Models** main window.

MIMO model configuration window is used to set up particular MIMO model. This window allow change parameters of the models and add/remove inputs, outputs and residuals.

View Selector at the top of the window (see Figure 27) are used to moving between **Context Panels** which represent different views of model's configuration.

Buttons Panel contains buttons which perform following functions:

- ◆ To add new input, click **Add Input** button. The new input will be added to inputs list on **Inputs** tab of the window.
- ◆ To add new output, click **Add Output** button. The new output will be added to outputs list on **Outputs** tab of the window.
- ◆ To add new residuum, click **Add Residuum** button. The new residuum will be added to residuals list on **Residuals** tab of the window.
- ◆ To remove input, output, residuum or model mark it on corresponding list and click **Remove selected** button,
- ◆ Close button closes the window.

Status bar at the bottom of the model's configuration window show MIMO model ID and name.

General panel

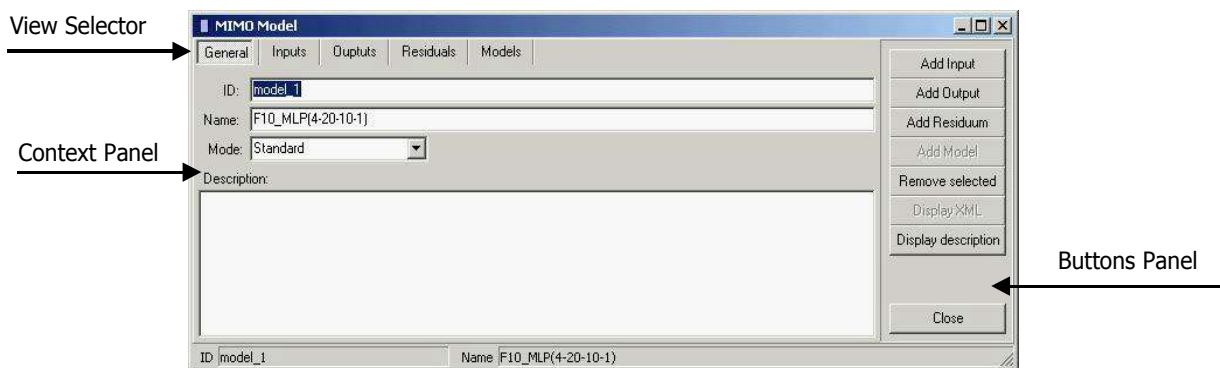


Figure 27: MIMO configuration window – General panel

General panel display and edit basic information of MIMO model, such as:

ID – unique model's ID,

Name – name of the model assigned by user,

Mode – current model's mode, it can be **Standard** (all model's inputs are taken from connected signals) and **MRO** – Model Reference Output (all AR inputs are taken from model's outputs),

Description – user description of the model,

Inputs panel

Inputs panel allow edit properties of each MIMO input (see Figure 28).

List box at the left side of **Inputs** panel is used to navigate between inputs. To configure particular input, click on corresponding list item, configuration will display on the left side of context panel.

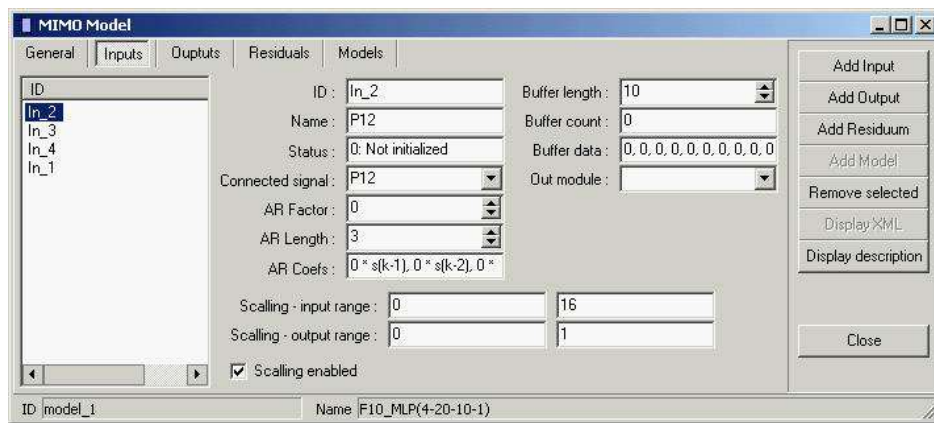


Figure 28: MIMO configuration window – Inputs panel

The following MIMO input's properties can be edited:

ID – unique input's ID,

Name – name of the input assigned by user,

Status – status of the input,

Connected signal – ID of the signal connected to the input (real time and historical values are taken from signal),

AR Factor, AR Length and AR coefs – parameters describes Auto Regressive model of the input (in fact it's AR model of signal),

Buffer length, Buffer count and Buffer data – parameters describes input's internal data buffer, length of the buffer, values count in buffer and its data,

Out module – properties used with MRO mode (see **General panel** description), it's name of output which should be taken as inputs of AR inputs,

Scaling - input range, Scaling - output range – data provided by connected signal will be scaled from input range to output range.

Scaling enabled – enable/disable scaling.

Outputs panel

Outputs panel allow edit prosperities of each MIMO output (see Figure 30).

List box at the left side of **Outputs panel** is used to navigate between outputs. To configure particular output, click on corresponding list item, configuration will display on the left side of context panel.

The following MIMO output's properties can be edited:

ID – unique output's ID,

Name – name of the output assigned by user,

Status – status of the output,

Connected signal – ID of the signal connected to the output (calculated model's output will be assigned to this signal),

Scaling - input range, Scaling - output range – data put in the connected signal will be scaled from input range (model's output domain) to output range (connected signal domain).

Scaling enabled – enable/disable scaling.

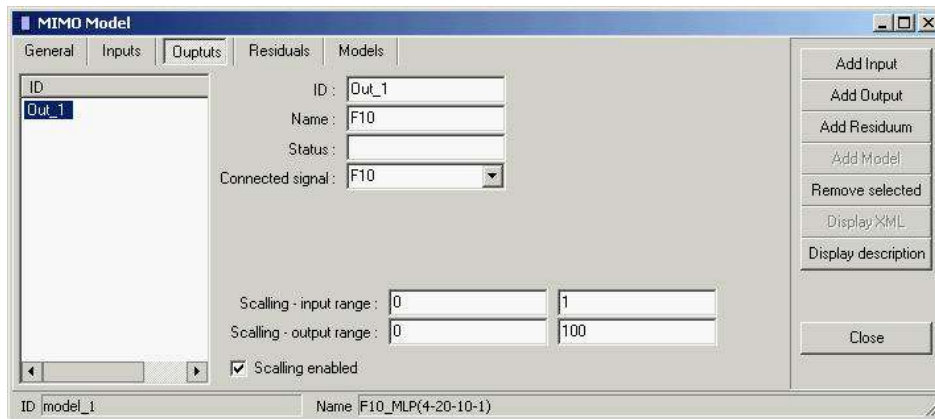


Figure 29: MIMO configuration window – Outputs panel

Residuals panel

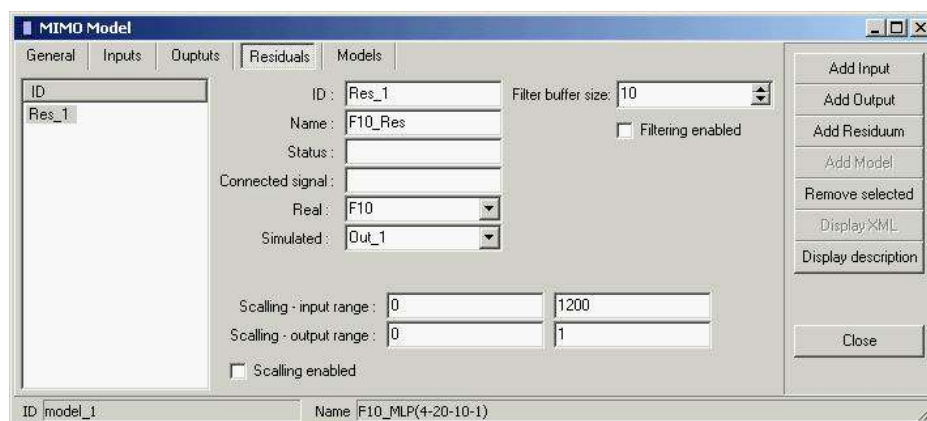


Figure 30: MIMO configuration window –Residuals panel

Residuals panel allow edit prosperities of each MIMO residual (see Figure 30).

List box at the left side of the panel is used to navigate between residuals. To configure particular residual, click on corresponding list item, configuration will display on the left side of context panel.

The following residual's properties can be edited:

ID – unique residual's ID,

Name – name of the residual assigned by user,

Status – status of the residual,

Connected signal – ID of the signal connected to the residual (calculated residual will be assigned to this signal),

Real – ID of the signal for which residual is calculated,

Simulated – ID of the model's output used to calculate residual,

Filter buffer size – size of residual filtering buffer,

Filtering enabled – enable/disable residual filtering,

Scaling - input range, Scaling - output range – data put in the connected Residual signal will be scaled from input range (residual's output domain) to output range (connected Residual signal domain),

Scaling enabled – enable/disable scaling.

Models panel



Figure 31: MIMO configuration window –Models panel

Model panel allow edit basic prosperities of each MISO models of MIMO model (see Figure 31).

List box at the left side of **Model panel** is used to navigate MISO models. To configure particular model, click on corresponding list item, configuration will display on the left side of context panel. More detailed prosperities, model plugin depended, is available by **Show details ...** button.

The following model's properties can be edited:

ID – unique model's ID,

Name – name of the model assigned by user,

Type – model's type (READ ONLY),

Status – status of the model (READ ONLY),

1.4.5 Configuring new MIMO model

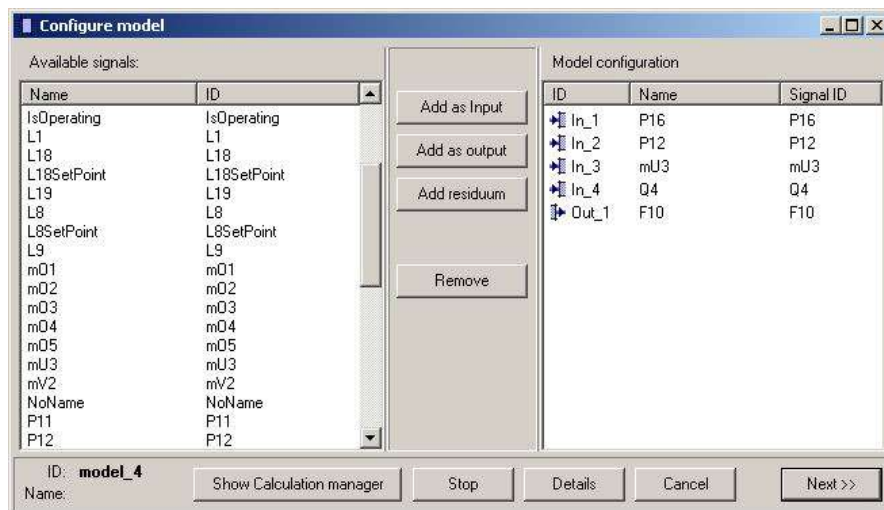


Figure 32: MIMO configuration window –Models panel

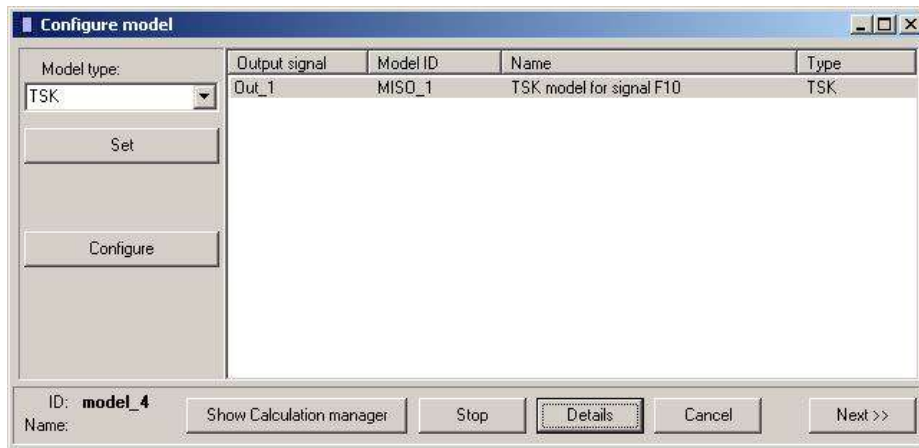


Figure 33: MIMO configuration window –Models panel

1.5 Verification

1.5.1 Invoking Verification window

To invoke Verification window, choose Modules | Verification from Main Window menu bar, or click Verification icon on Main Window toolbar.

1.5.2 Verification window

Verification window is used as a main window of Verification Module. The main purpose of this window is to provide environment to verification and validation estimated models.

The verifications of model is done in few steps. The first one is loading verification data (by default it is current data set). The second step is simulation of models, in Verification Module it's implemented as a plug-ins. Result of simulation can be compared on plot with original signal. Also some verification benchmarks are computed during simulation, it is presented in Details part of Verification Window.

The result of simulation can be viewed in two ways – plot view and details view. Plot view is used to compare result of simulation with original signal (see Figure 34). Details view, display some coefficients computed during simulation which are useful to value the model (see Figure 35).

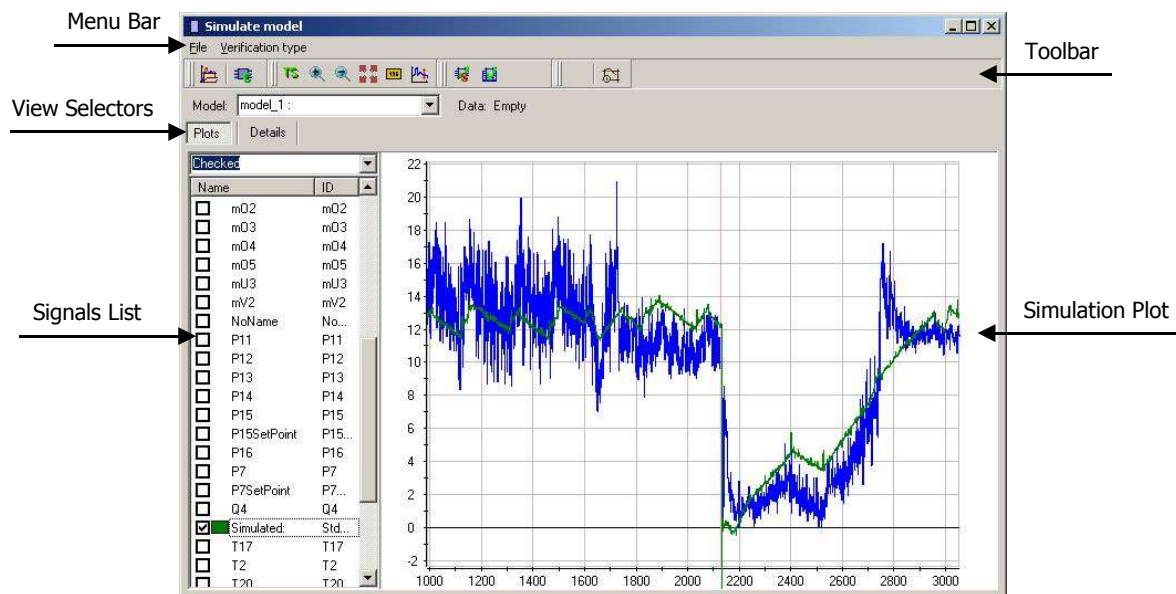


Figure 34: Verification window – plot view

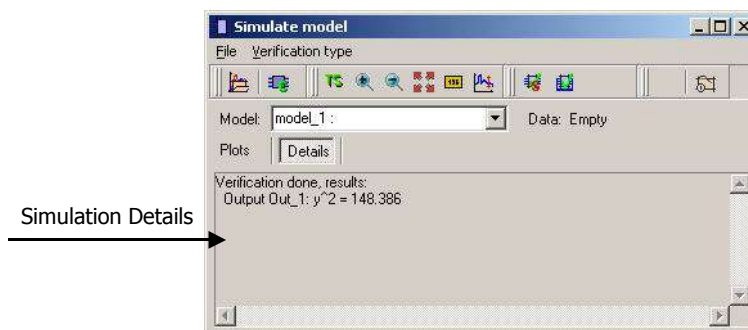


Figure 35: Verification window – details view

Model combo box allow to change currently simulated model.

Menu Bar

File Menu

- Load dataset – load verification data set from file in MITforRD data format,
- Close – close the window,

Verification type

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Toolbar

-  – load verification data,,
-  – invokes MIMO Model window,
-  – enable/disable showing time stamps on Plots view,
-  – zoom in time axis of Plots view,
-  – zoom out time axis of Plots view,
-  – undo zoom and full plot of Plots view,
-  – enable/disable showing of cursor position on Plots view,
-  – enable/disable plots normalize on Plots view,
-  – simulate current model,
-  – simulate current model using Model Reference Mode,
-  – close the window.

View Selector

View selector buttons are used to switch between details view and plot view. To change view click on corresponding button, Details button or Plots button.

Plot view is used to compare result of simulation with original signal (see Figure 34). Details view, display some coefficients computed during simulation which are useful to value the model (see Figure 35).

Signals List

Signal List is used to choose which signals should be plot on plots view. In the top of Signal List is combo box used to control list behavior, see Figure 36.

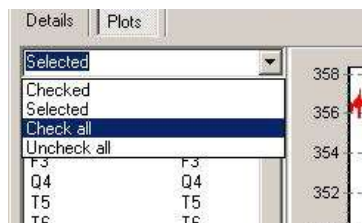


Figure 36: Signals List control combo box; Verification window - plot view

Changing behaviour of Signals List

- ◆ to change **Selected** list to **Checked** list (signals' list with check boxes), choose **Checked** item of combo; this option allow to plot multiple signals on the same plot by marking signal on the list;
- ◆ to change **Checked** list to **Selected** list, choose **Selected** item of combo; this option allow to plot only one signals on plot by choosing it from the list;
- ◆ **Check All** item is used to mark all signals on list (only when **Checked** option is chosen) – all signals are plotted;
- ◆ **Uncheck All** item allow to unmark all signals on list (only when **Checked** option is chosen) - clears plot area;

Simulated signals can be found at end of the list with name starting with **Simulated**: e.g. **Simulated: F10**.

Simulation Plot

To enable **Simulation Plot** view press **Plot** button on **View Selector**. The plot view is used to visualize results of simulation with ability to compare with original signal. On plot can be displayed one signals or multiple signals.

Simulation Details

To enable **Simulation Details** view press **Details** button on **View Selector**. Details view display some coefficients computed during simulation which are useful to value the model.

1.6 Plugins

1.6.1 Analysis plugins



1.6.1.1 Power Spectrum plug-in



– invokes Power spectrum plug-in window



– invokes Histogram plug-in window



– invokes Autocorrelation plug-in window



– invokes Trends plug-in window



– invokes Interpolating plug-in window



– close the window

To invoke Power Spectrum plug-in window, choose Analysis | Power spectrum from menu bar of Data Preprocessing window menu bar, or click Power spectrum icon on Data Preprocessing window toolbar.

The Power Spectrum plug-in allow compute and visualize on plot power spectrum and phase of currently selected signal. To compute spectrum and phase plug-in uses FFT transform.

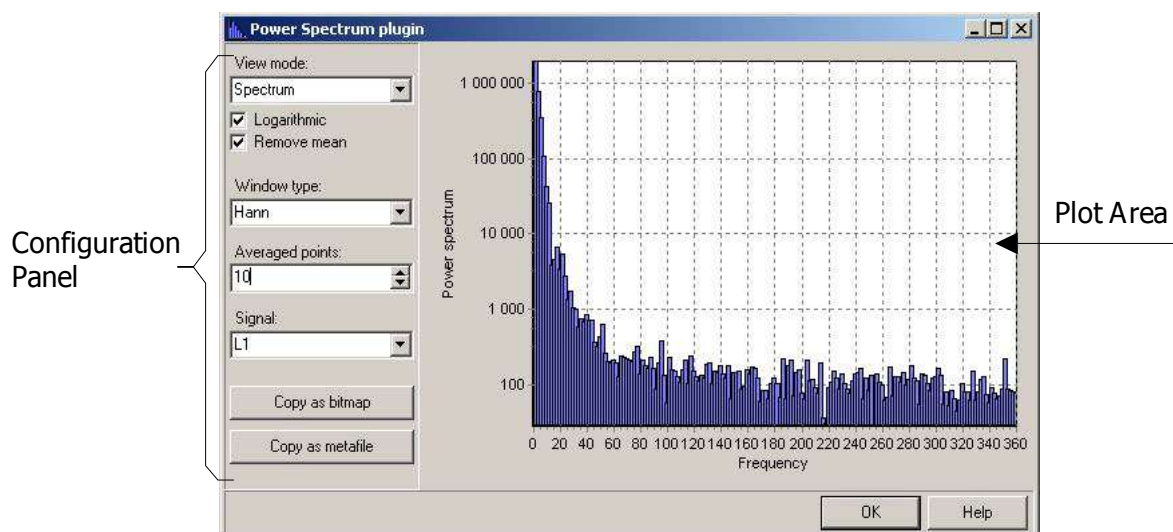


Figure 37: Power Spectrum plug-in window

Plot Area

Plot Area is used to display spectrum or phase of signal. Behaviour of plot can be changed using Configuration Panel. X-axis always show frequency. Values displayed on Y-axes depend on current View mode.

Configuration Panel

Configuration panel allow adjust functionality of Power Spectrum plug-in.

To change plot type use View mode combo box. The following plot types can be chosen:

- ◆ Spectrum – show spectrum plot, left Y-axis represent power,
- ◆ Phase – show phase plot, right Y-axis represent phase (in radians),

- ◆ **Both** – show spectrum and phase on the same plot, left Y-axis represent power and right one, phase.

Logarithmic check box allow switch spectrum Y-axis to logarithmic scale.

To remove mean during computation of spectrum or phase mark **Remove mean** check box.

Window type combo box is used to change FFT window type.

To change averaged points of computed spectrum or phase use **Averaged points** text box.

To change currently processed signal use **Signal** combo box.

Copy as bitmap button copy current plot to clipboard as bitmap.

Copy as metafile button copy current plot to clipboard as metafile.



1.6.1.2 Histogram plug-in

To invoke Histogram plug-in window, choose Analysis | Histogram from menu bar of Data Pre-processing window menu bar, or click Histogram icon on Data Preprocessing window toolbar.

The Histogram plug-in allow display on plot histogram of currently selected signal.

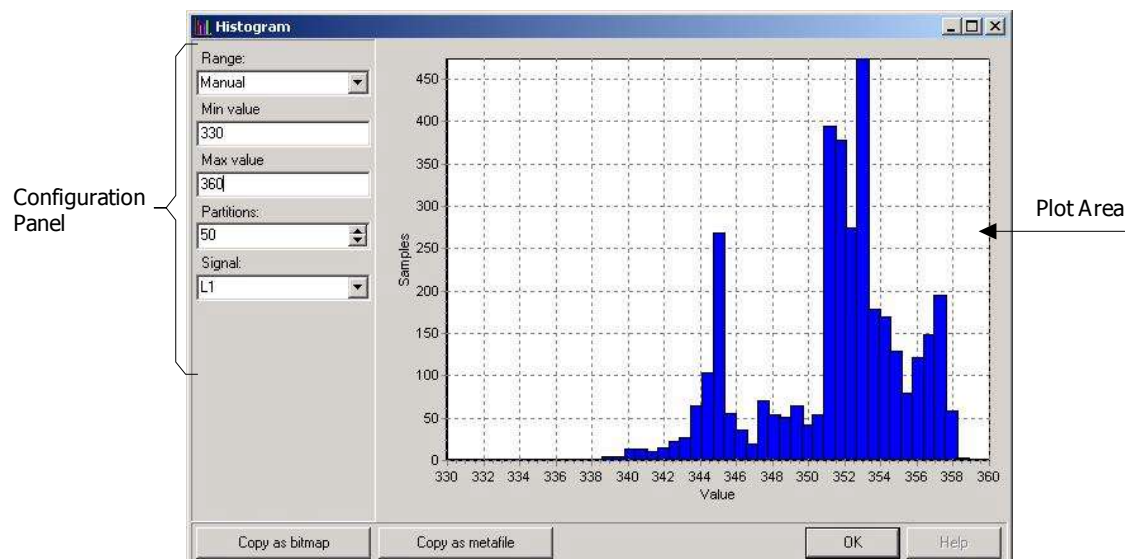


Figure 38: Histogram plug-in window

Plot Area

Plot Area is used to display histogram plot of signal. Behaviour of plot can be changed using Configuration Panel. X-axis always show signal domain, values displayed on Y-axes show samples count.

Configuration Panel

Configuration panel allow adjust behaviour of histogram.

To change way of choosing range of X-axis, use **Range** combo box. The following options are available:

- ◆ **Signal defined** – the range is taken from signal's configuration,
 - ◆ **Real** – as range of X-axis are taken real min and max value of signal,
- ◆ **Manual** – plot's range is taken from Min value and Max value text boxes of Configuration Panel.

Min value and Max value text boxes are used to manually provide ranges of X-axis.

To change partitions count of histogram use **Partitions** text box.

To change currently processed signal use **Signal** combo box.

Copy as bitmap button copy current plot to clipboard as bitmap.

Copy as metafile button copy current plot to clipboard as metafile.

1.6.2 Preprocessing plugins



1.6.2.1 Interpolating plug-in

Interpolating plug-in is used to complete missing sample's values.

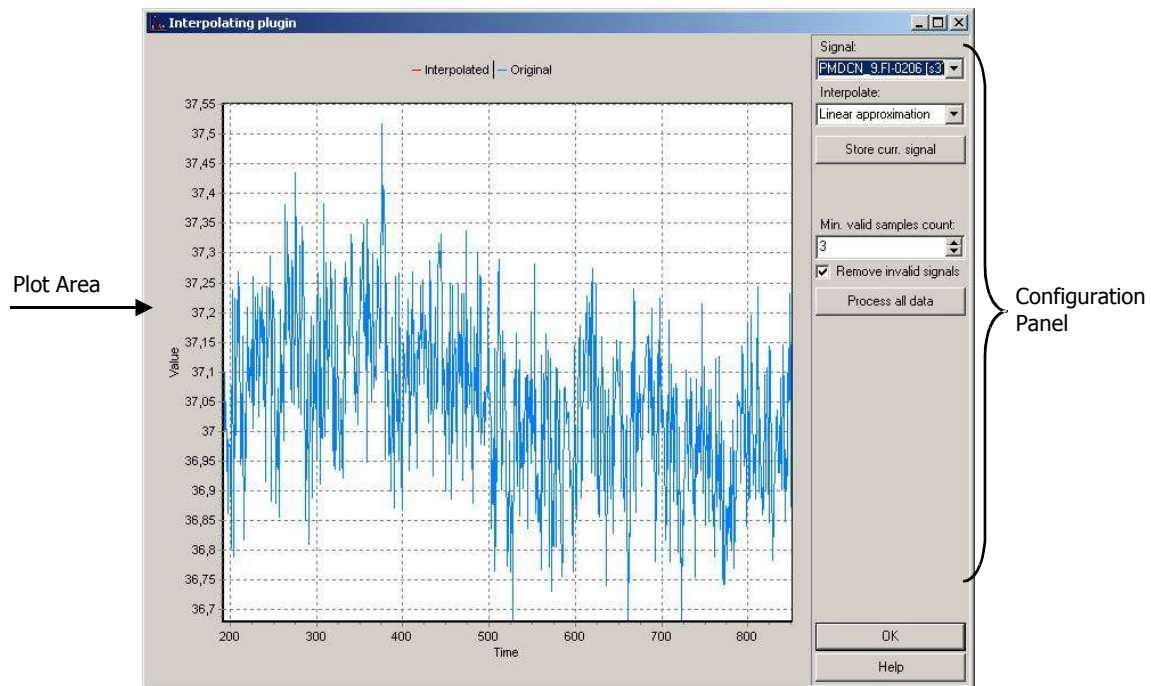


Figure 39: Interpolating plug-in window

Plot Area

Plot Area is used to compare current signal with signal after interpolation. X-axis always show sample number, values displayed on Y-axes show signal domain.

Configuration Panel

Configuration panel is used to interpolation management. To change interpolated signal use **Signal** combo box.

To change kind of interpolation, use **Interpolate** combo box. The following options are available:

- ◆ **Last value** – to complete missing values is used last good sample,
- ◆ **Linear approximation** – to complete missing values is linear approximation,

To store immediately interpolated signal press **Store curr. signal** button.

Min. valid samples count combo box is used to

Remove invalid signals enable/disable

To process all signal press **Process all data** button.

1.6.3 Import data plugins

1.6.3.1 Import text (csv) data

To invoke Import Data option, choose File | Import data from menu bar of Data Preprocessing window menu bar, or click Import data icon on Data Preprocessing window toolbar.

After choosing Import Data option Select import filter window will appear (see Figure 40). Filters are Import data plugins.



Figure 40: Import data – filter selection window

To choose Import text (csv) data filter, select it from Select filter combo box and click Ok.

Standard open dialog will appear, select file to import. The Import text file window will appear (see Figure 41).

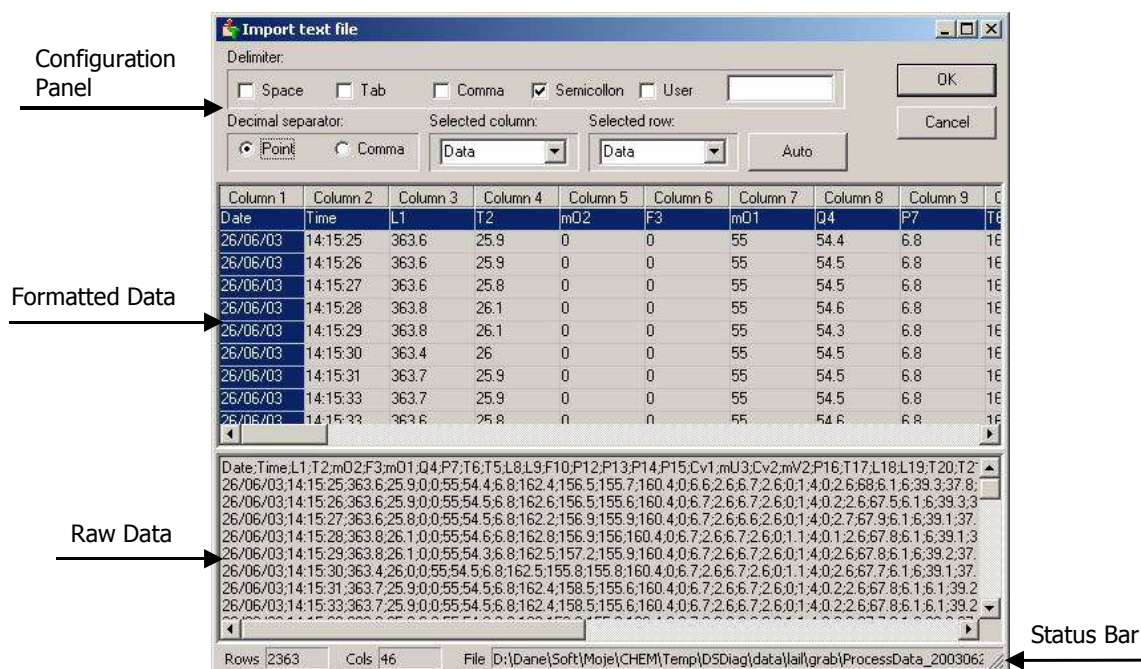


Figure 41: Import text file plug-in window

Import text file window is used to format raw data from txt file in such way that could translated to MITforRD internal data set representation. Formatting of data contains following activities:

- ◆ choosing data delimiters – character or characters which separate data,
- ◆ choosing decimal separator,
- ◆ marking types of column and rows – date, time and data,
- ◆ omitting of unused columns and rows.

Status Bar

Status Bar show basic information about imported data set. Each fields of status bar are described on Figure 20.

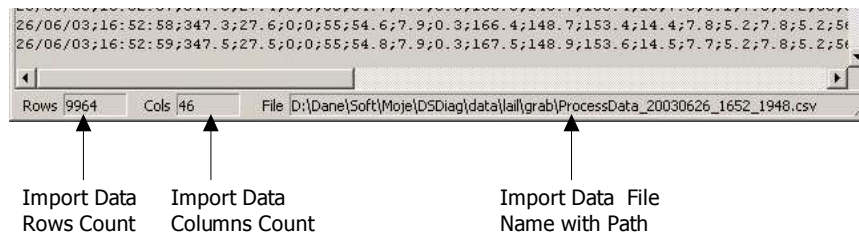


Figure 20: Status bar of Import text file plug-in window

Raw Data

Raw Data panel show part of original txt file. It is used just for information during formatting process.

Formatted Data

Formatted Data table show result of data formatting. If proper delimiter was chosen then table should contain in cell correct data and columns and rows count should be the same as in txt file.

Table's cells could contain values of imported data, time stamp or separated date and time, names of signals and so on.

Type of cell should be detected automatically but when it fails user can use Configuration Panel to do that.

Configuration Panel

Configuration Panel allow control process of formatting. **Delimiter** field is used to choose data separators – character or characters which separate data. To chose appropriate decimal separator use **Decimal separator** field.

Auto button can be used to automatic recognize imported file. This function automatic detect such parameters as decimal separator and data field delimiter.

Selected row and Selected Column allow choose column and row type.

The following rows' types are available:

- ◆ **Data** – a row consist data values,
- ◆ **Skip** – a row will be omitted,
- ◆ **ID's** – a row consist ID of signal,
- ◆ **Names** – a row consist name of signal,
- ◆ **Upper limit** – a row consist max value of signal (in most cases it is max of measure range),
- ◆ **Lower limit** – a row consist min value of signal (in most cases it is min of measure range)

The following columns' types are available:

- ◆ **Data** – a column consist data values,
- ◆ **Skip** – a column will be omitted,
- ◆ **Date** – a column consist time stamp date part,
- ◆ **Time** – a column consist time stamp time part,

1.6.4 Models plugins

1.6.4.1 Classic Linear Model plug-in

1.6.4.2 Polynomial Model plug-in

1.6.4.3 Multi-Layer Perceptron Model plug-in

1.6.4.4 Fuzzy Neural Network Model plug-in

UNDER CONSTRUCTION!

1.6.4.5 TSK Fuzzy Model plug-in

UNDER CONSTRUCTION!

1.6.4.6 Mendel - Wang Fuzzy Model plug-in

UNDER CONSTRUCTION!

2 MITforRD RTS

3 MITforRD RTS Config

4 Glossary

analytical redundancy - use of more than one not necessarily identical ways to determine a variable, where one way uses a process model.

DSS – an abbreviation of Decision Support System. The system, which is in the scope of the CHEM project.

DSS designer – the person who is responsible for the configuration of Decision Support System.

DSS operator – the person who is responsible for supervision and control of the process and is using DSS to fulfil that task. In the scope of this document, plant operator is person

fault – unpermitted deviation of at least one characteristic property or parameter of the system from acceptable / usual / standard conditions.

fault detection – an activity of determination of faults present in the system and time of detection.

fault isolation – an activity of determination of kind, location and time of detection of a fault. Follows fault detection.

FDI – an abbreviation for Fault Detection and Isolation.

FNN – an acronym for Fuzzy Neural Network,

general model – any functional, qualitative or quantitative representation of the system different then partial parametric model

GUI – an abbreviation for Graphical User Interface

historical process data – the data containing the recorded values of observable and calculated quantities in the process (e.g., *process variables, calculation variables, control variables, etc*).

identification – procedure of building mathematical model of a dynamic system based on measured data by adjusting parameters within a given model until its output coincides as well as possible (due to chosen criteria) with the measured output

identification data – raw data after preselecting process, ready for use by Identification Module

measured process variables – indicates that process variables coming from process.

MIMO – an abbreviation for Multi Input Multi Output.

MISO – an abbreviation for Multi Input Single Output.

MLP – an acronym for MultiLayer Perceptron a type of neural networks, also Multilayer Feed-forward Neural Network,

modelled process variables – in the scope of this documents the modelled process variables are the results of simulation carried out with use PPMs.

models management – all routines for model archive, visualization and comparison purposes.

parametric model – the description of the process in a form of equation with known structure and unknown parameters.

parameters estimation – procedure of adjusting parameters within a given model until its output coincides as well as possible (due to chosen criteria) with the measured output

partial model – the model of the part of the plant. All the plant will be divided into a smallest possible and separates parts, which can be modelled concerning available measurements (e.g. valve, super heater, evaporator).

partial parametric model – the model that is partial and parametric. Models describe relationships between measured signals. It is convenient to distinguish between input signals and output signals. The outputs are then partly determined by the inputs. In most cases, the outputs are also affected by more signals than the measured inputs. Such “unmeasured inputs” will be called disturbance signals or noise. If we denote inputs, outputs, and disturbances by u , y , and e , respectively, the relationship

can be depicted in the Fig. 8. All these signals are functions of time. In the identification context, only discrete-time points are considered, since the measurement equipment typically records the signals just at discrete-time instants. The modelling problem is then to describe how the three signals relate to each other.

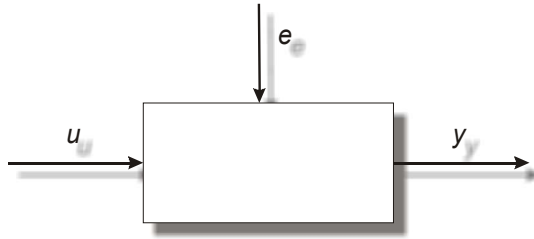


Figure 8: Input, Output and Disturbance

As a base TB3.5 is using Multi Input Single Output (MISO) model structure, where each input can be one of the following: *process inputs*, *disturbances* or *delayed process output*. Into MODEL IDENTIFICATION activity are implemented following basic types of models:

Fuzzy TSK models

Neural network (MLP)

Classic Fuzzy Model identified with Wang-Mendel method and its modifications:

Fuzzy Neural Network:

Such defined MISO models will be combined into a MIMO model, that is mentioned PPM. Besides MISO models into a PPM are additional objects responsible for data scaling, validation and external connections to the data source. Thus PPM provides common interface to any kind of implemented MISO models.

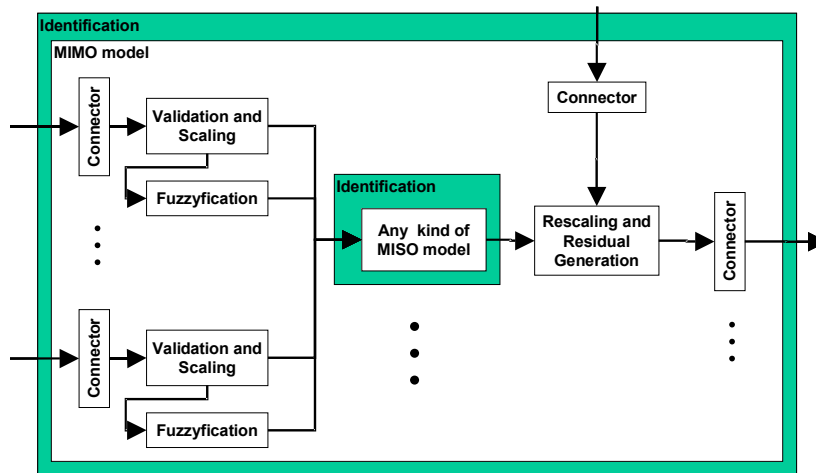


Figure 9: PPM internal structure.

PPM – an acronym for Partial Parametric Model.

PPM parameters – a set of variables (coefficients) that are adjusted during model identification in order to achieve maximum accuracy.

PPM skeleton – an object that represent PPM structure without parameters values.

PPM structure – the type and form of PPM equation with defined variables and parameters

process variables – in the scope of this document the process variables are the available measured values describing the process state.

raw data – historical values of process variables obtained from control system (DCS, SCADA) archives before any kind of processing

residuals to generate – A set of needed residuals from DSS

residuuum – a fault indicator - calculation variable which carries the information about fault existence. It is designed to have close to zero value when there is no fault in the system. The appearance of a fault causes that some of the residuals from the whole set of residuals have non-zero values. The residuals are usually calculated on the basis of **analytical redundancy** – as a difference between measured process variable and modelled one, *e.g., the difference between real flow through the valve and the flow calculated by the valve model.*

simulated process variables – the same as **modelled process variables**.

simulation – procedure of calculating PPMs outputs. Simulation algorithm can be chosen from the following:

Standard – in this case previous measured output values are used in prediction algorithm,

Model Reference Output – in this case only estimated values of output are applied for prediction.

structure definition – Selection of type and form of model equation, input and output signals and estimation criteria.

TSK – an acronym for Takagi-Sugeno-Kanga Fuzzy Model,

WMFM – an acronym for Classic Fuzzy Model identified with Wang-Mendel method,